LESSON 6.2

Division Expressions with Fractions and Mixed Numbers

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain what dividing a fraction by a fraction means, find the quotient with common denominators or by multiplying by the reciprocal, and check it by multiplying.

WHY THIS MATTERS Agent Rivera is dividing a 3/4-pound evidence bag into portions that each weigh 1/8 pound. She needs to know exactly how many portions she can make for testing. Can you solve the case?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Dividend *Spanish: Dividendo*	The total number being divided into equal groups (the number inside the division bracket).	In 8/9 ÷ 2/9, 8/9 is the dividend.
Divisor *Spanish: Divisor*	The number of equal groups you are dividing into (the number outside the division bracket).	In 8/9 ÷ 2/9, 2/9 is the divisor.
Reciprocal *Spanish: Recíproco*	A fraction turned upside down.	2/9 → 9/2
Quotient *Spanish: Cociente*	The answer when you divide.	8/9 ÷ 2/9 = 4
Simplify *Spanish: Simplificar*	To write a fraction with smaller numbers that names the same amount, like 2/4 = 1/2.	8/12 → GCF is 4 → 8÷4 / 12÷4 = 2/3
Mixed number *Spanish: Número mixto*	A whole number and a fraction written together, like 2 1/2. It must be rewritten as an improper fraction before dividing.	2 1/2 = 5/2

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A pitcher has 2/3 liter of juice. Each cup holds 1/6 liter. How many cups can be filled?

1. Write the division: 2/3 ÷ 1/6.
2. Flip the divisor and multiply: 2/3 × 6/1 = 12/3.
3. Simplify: 12/3 = 4. Check: 4 × 1/6 = 4/6 = 2/3.

Answer: 4 cups

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's try 1/2 ÷ 1/4. How many 1/4s fit in 1/2? Rename 1/2 as 2/4.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- To divide by _____ , I multiply by its reciprocal _____ .
- The quotient counts how many groups of _____ fit inside _____ .
- I did not need a common denominator because _____ .
- After multiplying across, I simplified _____ to _____ .

Word bank reciprocal | dividend | divisor | quotient | keep | change | flip | multiply across | simplify | groups | fit inside | tape diagram

WATCH OUT!

A common mistake

A common mistake in Divide Fractions is flipping the wrong fraction: students flip the dividend (the first fraction) instead of the divisor (the second fraction). For example, solving 3/4 ÷ 5/8 by flipping 3/4 to 4/3 and keeping 5/8 gives 4/3 × 5/8 = 20/24 = 5/6

LESSON 6.2

Division Expressions with Fractions and Mixed Numbers

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each division: will the quotient be GREATER than 1 or LESS than 1?

Groups: Quotient Greater Than 1 | Quotient Less Than 1

_____ 3/4 ÷ 1/4

_____ 5/6 ÷ 1/3

_____ 2/3 ÷ 1/6

_____ 1/4 ÷ 1/2

_____ 1/3 ÷ 2/3

_____ 2/5 ÷ 4/5

HINT Work out each quotient first (multiply by the reciprocal), then compare that quotient to 1.

PART 2 Practice the skill

show every step

2 **Circle the letter of the best answer. Show how you know.**

What is 1/2 ÷ 1/4?

(A) 1/8 (B) 1/2

(C) 2 (D) 4

HINT This asks how many 1/4-size pieces fit inside 1/2 — the pieces are smaller than the amount.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is 3/4 ÷ 1/8?

(A) 3/32
(B) 6
(C) 8/3
(D) 24

HINT How many 1/8-pieces fit into 3/4? Eighths are small, so expect several.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is 2/3 ÷ 1/3?

(A) 1/2
(B) 2
(C) 2/9
(D) 6

HINT Both fractions are thirds — you're counting how many 1/3-pieces sit inside 2/3.

SHOW YOUR WORK

PART 3 Apply it

real-world problems

5 **Circle the letter of the best answer. Show how you know.**

What is the reciprocal of 3/5?

(A) 1/3
(B) 3/5
(C) 5/3
(D) 5/5

HINT Re-read 'reciprocal' — it's about the two numbers of the fraction trading places.

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

6 **Circle the letter of the best answer. Show how you know.**

What is the first step in evaluating 2 1/2 ÷ 1/4?

(A) Rewrite 2 1/2 as the improper fraction 5/2

(B) Flip 2 1/2 to 1/2 2

(C) Divide 2 by 1/4 and ignore the half

(D) Add 2 + 1/2 + 1/4

HINT Which number in this problem is a mixed number?

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why does 5/6 ÷ 1/12 = 10? How does the bar model show this?

SENTENCE STARTER 5/6 ÷ 1/12 = 10 because 5/6 is the same as ___ twelfths, and ___ twelfths divided by 1 twelfth = ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.2

Division Expressions with Fractions and Mixed Numbers

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Divide each pair of fractions using the reciprocal shortcut.

Problem	Multiply by the reciprocal	Quotient
1/2 ÷ 1/6		
3/4 ÷ 1/4		
2/3 ÷ 1/3		
5/8 ÷ 1/4		

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each fraction division to its quotient.

_____ 4/5 ÷ 2/5
_____ 3/4 ÷ 3/8
_____ 5/6 ÷ 1/3
_____ 2/3 ÷ 1/6
_____ 7/8 ÷ 1/4
_____ 1/2 ÷ 3/4

(A) 2
(B) 2
(C) 2 1/2
(D) 4
(E) 3 1/2
(F) 2/3

3 **Mark and label each value on the number line.**

Place each quotient on the number line.



SHOW YOUR THINKING

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A recipe needs 2/3 cup of sugar. You only have a 1/6-cup scoop. How many scoops do you need?

(A) 2 scoops
(B) 3 scoops
(C) 4 scoops
(D) 6 scoops

Rewrite each fraction _____

Work it out _____

Simplify _____

5 **Circle the letter of the best answer. Show how you know.**

Agent Rivera filled 3/4 of an evidence bag, and that part weighs 3/8 pound. How much would a FULL bag weigh? (3/8 ÷ 3/4 asks how much is in one whole bag.)

(A) 9/32 pound
(B) 1/2 pound
(C) 2 pounds
(D) 1 1/8 pounds

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Problem** 2/3 ÷ 1/2
2. **Change ÷ to × (but forgot to flip)** 2/3 × 1/2
3. **Multiply** 2/6
4. **Simplify** 1/3

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why does 5/6 ÷ 1/12 = 10? How does the bar model show this?

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LESSON 6.2

Division Expressions with Fractions and Mixed Numbers

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Divide, then check your answer by multiplying: quotient × divisor should equal the dividend.

Problem	Quotient	Check: Quotient × Divisor
5/6 ÷ 1/3		
4/9 ÷ 2/3		
7/8 ÷ 1/2		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Division Error

1. **Problem** 2/5 ÷ 3/4
2. **Multiply by the reciprocal** 2/5 × 3/4
3. **Multiply** 6/20
4. **Simplify** 3/10

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Is 3/4 ÷ 1/2 the same as 1/2 ÷ 3/4? Solve both and explain why or why not. What does this tell you about the order in division?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Why does 5/6 ÷ 1/12 = 10? How does the bar model show this?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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